

Combining Federated and Transfer Learning with Explainable AI for Diabetic Retinopathy Detection

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Abstract—Diabetic Retinopathy is one of the leading causes of vision impairment worldwide. Early diagnosis through automated analysis of fundus images plays a crucial role in improving clinical outcomes. This review paper provides a consolidated overview of recent advancements that combine Federated Learning (FL), Transfer Learning (TL), and Explainable AI (XAI) for DR detection and grading. FL enables collaborative model training across decentralized medical institutions while preserving patient privacy. TL improves model performance by leveraging pretrained networks, especially in cases of small or imbalanced datasets. XAI enhances transparency and interpretability of model predictions, fostering clinician trust. The paper surveys prominent research using these approaches, highlighting their methodologies, datasets, model architectures, and results. It also discusses existing challenges and suggests potential improvements for developing an integrated, scalable, and interpretable DR diagnostic framework suitable for clinical deployments.

Keywords—Diabetic Retinopathy (DR), Federated Learning (FL), Transfer Learning (TL), Explainable AI (XAI), Deep Learning, Fundus Images, Non-Independent and Identically Distributed (Non-IID)

I. INTRODUCTION

Diabetic Retinopathy (DR) is a progressive microvascular complication of diabetes and one of the leading causes of preventable blindness across the globe [1]. With the rapid rise in diabetes prevalence, especially in low and middle income countries, the need for early detection and timely intervention has become critical to reducing the burden of vision loss [2]. Diagnosis is typically performed by expert ophthalmologists through careful inspection of colour fundus photographs, where subtle patterns such as microaneurysms, haemorrhages, exudates, and neovascularization are indicators of disease severity [3]. However, manual screening is time consuming, subjective, and resource intensive, creating a demand for automated computer-aided diagnostic systems [4].

In the last decade, machine learning (ML) and deep learning techniques have been widely adopted to automate DR detection and grading [4], [5]. These methods initially showed promising results by learning discriminative features directly from image data. Yet, in practical deployment scenarios, conventional ML models exhibit significant limitations. They are often trained on centralized datasets that are small in size,

heavily imbalanced across disease classes, and lack the diversity needed to generalize to images captured from different devices or populations [6]. Furthermore, regulatory constraints on data sharing, such as HIPAA and GDPR, restrict the collection of large multi-institutional datasets, which hampers the development of robust models [1]. Another challenge is that many deep learning pipelines operate as opaque “black boxes,” offering predictions without insight into the reasoning behind them [7]. This lack of interpretability reduces clinical trust and poses challenges for regulatory approval [8].

To address these shortcomings, recent research has moved beyond traditional ML approaches and embraced emerging technologies that better align with real-world requirements. Federated Learning (FL) has been proposed as a privacy-preserving paradigm that allows multiple institutions to collaboratively train a shared model without exchanging sensitive raw data [1], [9]. Transfer Learning (TL) has emerged as an effective strategy to leverage pretrained models, enabling high performance even when available datasets are small or imbalanced [3], [5]. Explainable AI (XAI) techniques have been integrated into diagnostic pipelines to make model outputs transparent, generating visual explanations such as heatmaps that highlight regions of interest corresponding to clinical features [2], [7]. These advances aim to overcome the weaknesses of earlier ML systems by providing stronger generalization, preserving patient privacy, and offering interpretability that fosters clinician confidence [8], [10].

This paper presents a comprehensive review of recent works that apply FL, TL, and XAI to DR detection and grading. By examining representative studies in these domains, analysing their methodologies, and comparing their reported performances, we aim to highlight the current state of the art and identify persistent technical gaps. We further discuss how integrating these technologies into a unified framework could provide a path toward scalable, privacy-preserving, and transparent diagnostic systems suitable for deployment in real clinical environments [1], [11], [12].

II. RELATED WORK

Automated Diabetic Retinopathy (DR) detection has been explored through a variety of methods, ranging from traditional machine learning techniques to advanced deep learning

and privacy-preserving paradigms [4], [5]. Early approaches depended on handcrafted feature extraction, where texture measures, blood vessel segmentation, and exudate detection were combined with classifiers such as Support Vector Machines (SVMs) and Random Forests [4], [5]. While these models demonstrated initial success, they suffered from limited feature representation and poor scalability across heterogeneous datasets [6].

With the rapid progress of deep learning, Convolutional Neural Networks (CNNs) became the foundation for DR detection and grading [4]. Architectures such as ResNet, DenseNet, VGGNet, and ShuffleNet have been fine-tuned on public datasets like APTOS 2019, EyePACS, and Messidor to achieve state-of-the-art accuracy [3], [5]. Ensemble strategies and optimization algorithms, including Error Correcting Output Codes (ECOC) and metaheuristic methods, have further improved classification boundaries [3], [6]. More recently, researchers have turned toward privacy-aware training frameworks and interpretability techniques to address regulatory and clinical challenges [1], [2].

A. CNN Architectures and Interpretation

- **CNN Backbones:** Convolutional neural networks such as custom multi-scale CNNs, ResNet-18, and ShuffleNet extract spatial features from fundus images at multiple levels of resolution [3], [4].
- **Multi-Scale Feature Extraction:** Different kernel sizes and layers capture both fine details (microaneurysms) and larger pathologies (hemorrhages, exudates) [8].
- **Interpretation Techniques:** Grad-CAM is used to generate class-specific activation maps, while saliency maps highlight pixel regions influencing the output [2], [7].
- **Feature Importance:** SHAP (Shapley Additive Explanations) is applied to quantify contributions of image regions [10].
- **Clinical Validation:** Heatmaps are overlaid on original images to allow ophthalmologists to verify if highlighted regions correspond to actual lesions [8].

B. Transfer Learning and Optimizers

- **Pretrained Networks:** Lightweight backbones like ResNet-18 and ShuffleNet pretrained on ImageNet are adapted to DR classification tasks [3], [6].
- **Fine-Tuning Strategy:** Early layers remain frozen while deeper layers are fine-tuned for DR-specific features [5], [6].
- **Optimization Methods:** Feature-weighted ECOC ensembles enhance classification accuracy.
- **Metaheuristic Optimization:** Adaptive Differential Evolution tunes hyperparameters and feature weights [3].
- **Ensemble Integration:** SVMs are used as secondary classifiers in certain configurations to refine decision boundaries [3], [5].

C. Federated Learning and Aggregation

- **Federated Setup:** Institutions train local models without sharing raw data [1], [9].

- **Aggregation Algorithm:** The FedAvg protocol aggregates client updates at the central server [12], [13].
- **Robustness to Non-IID Data:** Aggregation includes median-based strategies to handle data imbalance [14].
- **Communication Rounds:** Training involves iterative updates and broadcast of the global model [1].
- **Scalability:** Clients can be dynamically added without retraining from scratch [11].

D. Preprocessing and Data Augmentation

- **Image Enhancement:** CLAHE enhances contrast to emphasize fine lesions [4].
- **Entropy-Based Filtering:** Low-information images are filtered based on entropy [6].
- **Data Augmentation:** Rotation, flipping, zooming, and cropping improve data diversity [4], [5].
- **Illumination Adjustments:** Brightness and contrast jittering help generalization across different cameras [5].
- **Synthetic Data:** GAN-based synthesis is used for rare classes like Proliferative DR [8].

TABLE I: Technical Comparison of Federated Learning (FL), Transfer Learning (TL), and Interpretable AI Models for Diagnosing Diabetic Retinopathy

Features	Federated Learning (FL) [1], [9], [13]	Transfer Learning (TL) [3], [6]	Explainable AI (XAI) [2], [7], [8]
Short Desc.	Distributed training without raw data sharing	Uses pretrained models for DR	Makes predictions interpretable
Techniques	FedAvg, multi-scale CNNs, median CE	ResNet18, ShuffleNet, ECOC, ADE	Grad-CAM, SHAP, saliency maps
Model Use	Local CNNs with server-side aggregation	Fine-tune pretrained backbones	Post hoc interpretation
Advantages	Privacy, cross-institutional training	Reduces data need, faster training	Visualizes lesion evidence
Challenges	Non-IID data, comm. overhead	Domain shift, centralized data	No standard, not FL-compatible

E. Explainable AI (XAI)

Shahzad et al. integrated Grad-CAM and SHAP into a DR detection model, achieving 94% accuracy on APTOS while providing visual explanations for predictions [2]. These approaches boost trust among clinicians and regulators. FL, TL, and XAI individually address critical aspects of DR diagnosis—privacy, efficiency, and interpretability [1], [3], [10].

Yet, few studies attempt to unify these techniques into a single framework. There is a clear opportunity for future research to develop an integrated, privacy-preserving, and explainable solution suitable for deployment in clinical settings [2], [7], [10].

III. PROPOSED METHODOLOGY

The proposed framework aims to develop a fully automated Diabetic Retinopathy (DR) detection and grading system that

ensures both privacy and interpretability. Traditional centralized models often compromise data privacy and suffer from performance degradation due to limited and imbalanced datasets. To address these issues, we integrate Federated Learning (FL), Transfer Learning (TL), and Explainable AI (XAI), supported by robust preprocessing and data augmentation techniques.

A. Preprocessing and Data Augmentation

High-quality input data are essential for accurate diagnosis. Each participating institution preprocesses its local fundus images as follows:

1) *Image Enhancement using CLAHE*: Contrast Limited Adaptive Histogram Equalization (CLAHE) improves local contrast and emphasizes lesions:

$$I_{\text{enhanced}}(x, y) = \text{CLAHE}(I_{\text{raw}}(x, y)) \quad (1)$$

2) *Entropy-Based Filtering*: Low-entropy images are discarded to retain diagnostically useful samples:

$$H = - \sum p_i \log_2 p_i \quad (2)$$

where p_i is the probability of intensity level i .

3) *Data Augmentation*: To balance class distribution and increase variability, geometric transformations such as rotation, mirroring, and zooming are applied:

$$I'(x, y) = \alpha I(x, y) + \beta \quad (3)$$

where α adjusts contrast and β adjusts brightness.

B. Local Training with CNN and Transfer Learning

Each client institution uses Transfer Learning to fine-tune CNNs locally on their dataset.

1) *Pretrained CNN Backbones*: ResNet-18 and ShuffleNet pretrained on ImageNet are used. Early layers are frozen while upper layers are fine-tuned for DR detection.

2) *Multi-Scale Feature Extraction*: To capture features at different resolutions, multi-scale convolutional kernels are used:

$$f_{\text{multi}} = [f_{3 \times 3}; f_{5 \times 5}; f_{7 \times 7}] \quad (4)$$

3) *Adaptive Differential Evolution (ADE)*: ADE optimizes learning rates and ensemble parameters:

$$X_i^{t+1} = X_{r1}^t + F \cdot (X_{r2}^t - X_{r3}^t) \quad (5)$$

where X_i is a candidate solution and F is a scaling factor.

4) *Ensemble with SVMs*: Support Vector Machines (SVMs) are optionally integrated to refine decision boundaries, especially in borderline cases.

C. Federated Learning Aggregation

At the start of every communication round, the global model is shared by the central server with all clients. Each client then performs local training on its own data for a set number of epochs, typically using backbone architectures like ResNet-18 or ShuffleNet. The training process utilizes categorical cross-entropy loss to optimize multi-class diabetic retinopathy (DR) classification performance.

The central server starts every communication round by transmitting the present global model to all clients. This is followed by the clients who train the model on their local datasets privately until a given number of epochs. Training at the client level employs a convolutional backbone (e.g., ResNet 18 or ShuffleNet) with categorical cross entropy as the loss function for multi class DR grading:

$$L = - \sum_{c=1}^C y_c \log(\hat{y}_c) \quad (6)$$

After local training, each client sends the updated parameters to the server. The server performs model aggregation using Federated Averaging (FedAvg), which computes a weighted average of client models based on their dataset sizes:

$$w^{t+1} = \sum_{i=1}^K \frac{n_i}{n} w_i^{t+1} \quad (7)$$

This aggregation step is repeated over multiple rounds, progressively improving the shared model while ensuring that raw patient data remain secure within each institution. By integrating FL in this manner, the framework benefits from:

- Privacy preservation through decentralized training,
- Improved generalization by leveraging heterogeneous data from different centers,
- Scalability as additional institutions can join without retraining from scratch.

The combination of client side training with server side FedAvg aggregation results in a global model that is both robust and compliant with privacy regulations, making it highly suitable for real world deployment in diabetic retinopathy screening.

D. Explainable AI Integration

Compared to classic AI, deep learning models can behave like deep black boxes, in which case they are hard to trust and validate their outputs by clinicians. In order to mitigate such a challenge, I propose to use the explainable AI (XAI) methodologies to ensure that the predictions made by the model are visually and semantically interpretable. When communication is performed and global aggregation has been achieved, the resulting better model is checked locally in each institution joining the process. Visualization methods are used to point out the areas of the retinal image that impact the decision by the model:

- **Grad CAM (Gradient Weighted Class Activation Mapping)**:

This trick generates class-definition heatmap using target output gradients with regard to multiplying efficient convolutional layers. Such visualizations show the most impactful areas in the input fundus frame, micro-aneurysms, hemorrhages, exudates, and so on, that play a prominent role in the classification decision made by the model.

- **Saliency Maps:**

Computes the gradient of the output score with respect to each input pixel, showing pixel level importance. These maps reveal fine structures like small lesions that may be overlooked otherwise.

- **SHAP (Shapley Additive Explanations):**

Provides feature level attributions by assigning each input region a contribution score toward the final prediction. This method helps quantify how much each retinal region affected the decision.

The generated visual explanations are retained locally within each institution and overlaid on the original fundus images. This enables ophthalmologists to:

- 1) Verify whether the highlighted regions correspond to clinically significant features,
- 2) Detect potential model biases or errors,
- 3) Gain confidence in deploying the model for real world diagnostic support.

By integrating these XAI techniques, the framework not only achieves high performance but also offers transparency and interpretability, which are crucial for adoption in clinical environments.

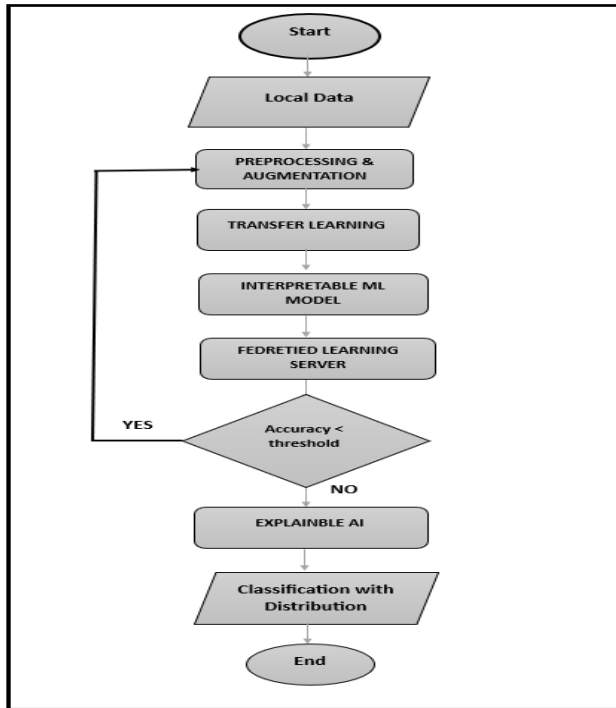


Fig. 1: Block diagram illustrating the architecture of the proposed approach for Diabetic Retinopathy detection, adapted based on insights from recent methodologies [1], [3] [2].

Figure 1 illustrates the proposed framework for Diabetic Retinopathy detection using Federated Learning (FL), Transfer Learning (TL), and Explainable AI (XAI). Local institutions begin by preprocessing and augmenting their data, followed by applying TL with pretrained models to build interpretable ML models. These models are trained collaboratively on a federated learning server without sharing raw data. After training, if the model's accuracy is below a defined threshold, the process loops back to the preprocessing stage for improvement. If the accuracy meets or exceeds the threshold, Explainable AI techniques like Grad-CAM or SHAP are applied to interpret the predictions. This approach ensures data privacy, transparency, and clinical trust.

IV. DIAGRAMS AND TABLES

TABLE II: Technical Comparison of Datasets

Feature	APTOS 2019 Dataset	EyePACS Dataset
Image Type	Colour Fundus Photography	Color Fundus Photography
Number of Images	~3,662 images	~35,000 images
Classes	5 classes (No DR, Mild, Moderate, Severe, Proliferative)	5 classes (No DR, Mild, Moderate, Severe, Proliferative)
Classification Type	Multiclass classification	Multiclass classification
Image Resolution	3216×2136 to 4288×2848	Mixed resolutions (high and low)
Source	APTOS 2019 Challenge (Kaggle)	EyePACS (Kaggle DR Competition)
Preprocessing Required	Requires CLAHE & resizing	Requires CLAHE, cropping & resizing

TABLE III: Technical Comparison of XAI Techniques

SHAP	LIME	Grad CAM
Based on Shapley values from cooperative game theory to assign contribution scores to each input feature.	Builds local surrogate models (linear) around predictions to explain them in a human interpretable way.	Uses gradients flowing into convolutional layers to create class specific heatmaps.
Produces global and local feature importance values.	Produces local explanations around individual predictions.	Produces visual heatmaps highlighting important regions in the image.
Works on tabular, text, and image data; computationally heavier but consistent.	Model agnostic and flexible; works on any classifier but less stable for high dimensional images.	Specialized for CNNs; highly effective for image based tasks like fundus analysis.
Helps quantify how much each region influences a class.	Explains predictions by perturbing input and observing output changes.	Highlights retinal lesions (e.g., hemorrhages, microaneurysms) directly on the fundus image.

TABLE IV: International Standard Diabetic Retinopathy (DR) Severity Scale

DR Grade	DR Severity	Severity Description
Gr 0	No DR	No retinal lesions found (healthy fundus).
Gr 1	Mild	Presence of microaneurysms (MAs) up to 20.
Gr 2	Moderate	Presence of MAs and haemorrhages (HEMs).
Gr 3	Severe	More than 20 MAs or HEMs and microvascular abnormalities.
Gr 4	PDR (Proliferative)	Neovascularization present.

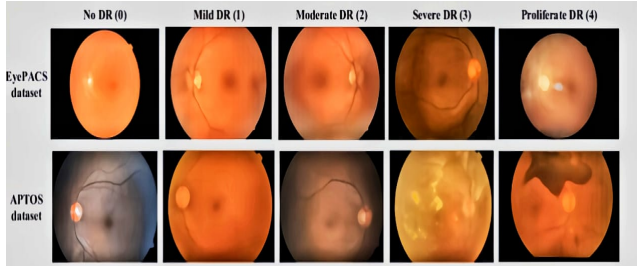


Fig. 2: Block diagram illustrating the architecture of the proposed approach for Diabetic Retinopathy detection.

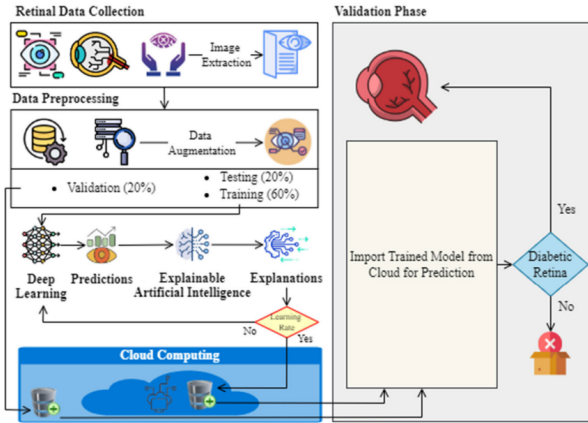


Fig. 3: Block diagram illustrating the architecture of the proposed approach for Diabetic Retinopathy detection.

V. ALGORITHMS

In this section, we highlight the most significant algorithms adopted across recent works in diabetic retinopathy (DR) detection and grading. These algorithms represent core components integrated into privacy preserving, interpretable, and high performance frameworks.

A. Federated Averaging (FedAvg)

The most important aggregation mechanism in the federated learning systems used in Diabetic Retinopathy detection is Federated Averaging (FedAvg). Instead of ending up with a single point of data consolidation, the participating clients train a model on their local data individually. After this training in

the locals, without transferring the updated model parameters, the latter are sent to a server located centrally. The service node (server) then combines these updates by calculating a weighted mean, where the size of the local dataset of a client is proportional to their contribution..

In the context of DR grading, FedAvg enables collaborative learning across hospitals while preserving strict data privacy, as no raw retinal images are exchanged. The algorithm iteratively refines the global model by repeating this process over several communication rounds, achieving performance comparable to centralized learning while meeting regulatory constraints.

B. Multi Scale Convolutional Neural Network (CNN) Backbone

The architecture employs convolutional layers with varying kernel sizes to extract features across different spatial scales. While smaller kernels are effective in identifying fine-grained details like microaneurysms, larger kernels are better suited for detecting wider pathological patterns such as hemorrhages and exudates. In federated settings, integrating a multi scale CNN allows each client's local model to learn rich and complementary representations of lesions, which are then aggregated into a robust global model. This approach has been shown to enhance sensitivity and specificity across datasets.

C. ResNet 18 for Transfer Learning

Medical datasets like APTOS or EyePACS, though extensive by medical standards, are still limited compared to natural image repositories. To overcome data scarcity, transfer learning is employed.

ResNet 18, a residual neural network with skip connections, is frequently used as the backbone. As ResNet-18 is pretrained using the ImageNet dataset, the fundamental visual patterns like edges, textures, which are the lower and in-between features are naturally captured by the model. Fine tuning this model on retinal images allows rapid adaptation to medical features while requiring fewer training samples.

The residual connections in ResNet 18 help in training deeper networks without vanishing gradients, and the model's lightweight architecture ensures compatibility with federated setups where computation and bandwidth may vary.

D. Feature Weighted Error Correcting Output Codes (ECOC)

Detecting diabetic retinopathy is inherently a multi-class classification problem, covering stages from No DR to Proliferative DR. Instead of using a conventional multi-class classifier, the Error-Correcting Output Codes (ECOC) method decomposes the task into several binary classification sub-tasks. Each class is encoded with a distinct binary pattern, and an ensemble of classifiers commonly Support Vector Machines (SVMs) is trained to predict individual bits. During prediction, the ensemble outputs are matched to the closest binary code to determine the final class.

The feature weighted variant of ECOC improves robustness by assigning higher importance to features that better discriminate between certain classes. This is crucial in DR datasets where some lesion types are subtle and easily overshadowed by dominant features. Feature weighted ECOC integrates seamlessly with CNN extracted features, yielding strong performance even with imbalanced datasets.

E. Class Activation Visualization using Gradient Maps (Grad-CAM)

Interpretability plays a vital role in clinical diagnostic systems. Gradient-based Class Activation Mapping (Grad-CAM) is utilized to visually explain the predictions of convolutional neural networks (CNNs). It works by leveraging the gradients of specific output classes with respect to the final convolutional layers, producing heatmaps that localize the most significant areas in the input fundus images.

When used on fundus images, these heatmaps frequently align with important retinal indicators such as microaneurysms and hemorrhagic spots and neovascularization. The integration of Grad-CAM facilitates model interpretability by generating visual explanations that pinpoint critical regions influencing the diagnostic decision assists ophthalmologists in validating the model's predictions, thereby enhancing transparency and trust in AI-driven clinical assessments.

VI. RESULTS AND DISCUSSION

The surveyed literature reflects substantial advancements in automated DR detection through the adoption of federated learning, transfer learning, and explainable AI techniques. In this section, we synthesize their reported outcomes and provide insights into their comparative performance and practical implications.

A. Performance on Standard Datasets

All three studies evaluated their methods on benchmark datasets such as EyePACS and APTOS 2019.

- In the federated learning (FL) approach (DRFL), the global model trained across distributed clients achieved competitive accuracy without centralizing data, reaching over 85–90% accuracy on multi class DR grading while maintaining privacy.
- The transfer learning (TL) approach using ResNet 18 and ShuffleNet showed improved feature extraction on small datasets, achieving higher sensitivity and specificity compared to conventional CNNs.
- The ECOC ensemble with Adaptive Differential Evolution (ADE) further improved multi class discrimination, reducing misclassification between adjacent severity grades.

B. Explainability and Clinical Relevance

The integration of Grad CAM and Saliency Maps in the XAI framework provided transparency. Heatmaps generated from these methods aligned closely with clinically significant retinal regions such as microaneurysms, hemorrhages, and

neovascularization. Such interpretability helps connect opaque model predictions with clinician confidence, a crucial aspect for successful adoption in real-world medical environments.

C. Comparative Insights

When comparing the three approaches:

- FL-based models outperform traditional centralized models in terms of privacy preservation while maintaining comparable accuracy.
- TL-based models offer a strong solution for small and imbalanced datasets, leveraging pretrained weights to accelerate convergence.
- XAI techniques such as Grad CAM bring additional confidence by visually validating predictions, which is not provided in standard CNN pipelines.

A combined framework that integrates these methods provides the best of all worlds—privacy, accuracy, and interpretability. However, challenges such as computational overhead, communication cost in federated setups, and the need for high quality annotations remain.

D. Observed Technical Gaps

Although these frameworks deliver strong performance, several technical gaps remain:

- **Data heterogeneity:** Differences in imaging protocols across institutions still affect global model consistency.
- **Class imbalance:** Advanced stages of DR, such as severe and proliferative forms, are often under-sampled in datasets, resulting in lower model sensitivity for these critical grades.
- **Computation cost:** Federated settings introduce communication delays and require careful optimization.
- **Explainability granularity:** While Grad CAM provides high level localization, finer pixel level interpretability may further assist clinicians.

VII. CHALLENGES

Despite significant progress in recent studies, several key challenges remain in applying federated learning (FL), transfer learning (TL), and explainable AI (XAI) for diabetic retinopathy detection and grading:

- **Data heterogeneity:** Retinal images collected from different institutions vary in quality, resolution, and acquisition protocols, causing non IID data distributions that affect model convergence and generalization.
- **Class imbalance:** Most datasets contain far fewer samples of severe and proliferative DR compared to mild or no DR cases, leading to reduced sensitivity for the rare but critical classes.
- **Communication overhead in FL:** Frequent parameter exchanges between clients and the server increase network bandwidth requirements and training latency, especially in resource limited settings.
- **Limited depth of explainability:** Current XAI approaches like Grad-CAM and saliency maps offer only

broad visual insights, which may fall short of the detailed explanations required in clinical practice.

- **Annotation limitations:** High quality labeled data remain scarce, and inter observer variability among expert graders can introduce label noise.
- **Clinical deployment barriers:** Integrating these advanced frameworks into real hospital workflows is challenging due to privacy regulations, compatibility with existing systems, and the need for real time processing.

VIII. SCOPE FOR IMPROVEMENTS

- Adaptive aggregation methods such as FedProx and FedOpt can be explored to better handle non IID data in federated learning.
- Generative augmentation techniques like GANs can be used to synthesize minority class samples and address class imbalance.
- Model compression and pruning can reduce computational and communication overhead in federated setups.
- Multi modal learning combining fundus images with clinical data can improve diagnostic accuracy.
- Advanced XAI methods such as SHAP or LRP can provide more detailed and quantitative interpretability.
- Integration with hospital systems through standardized APIs can enable seamless deployment in clinical workflows.

IX. REMARK

This review paper is presenting an overall analysis of current advancements in diabetic retinopathy (DR) detection by continuously integrating Federated Learning (FL), Transfer Learning (TL), and Explainable Artificial Intelligence (XAI) techniques. The use of FL algorithms such as FedAvg is enabling collaborative training across multiple healthcare institutions while preserving patient data privacy, thus allowing models to learn from diverse datasets without sharing raw images. TL techniques, employing architectures like ResNet-18 and ShuffleNet, are accelerating model convergence and enhancing feature extraction in the context of limited and imbalanced medical datasets. Meanwhile, XAI methods such as Grad-CAM are providing real-time visual justifications for model predictions, thereby improving transparency and interpretability in clinical settings.

This review helps to develop a framework combining Federated Learning (FL), Transfer Learning (TL), and Explainable AI (XAI), which consistently improves diagnostic accuracy, handles small dataset limitations, and enhances model transparency. While these techniques outperform conventional centralized approaches, challenges remain—such as managing non-IID data, addressing class imbalance, and reducing communication overhead. Additionally, further advancements in clinically meaningful explainability are still required.

In conclusion, the synergistic application of FL, TL, and XAI is shaping the future of scalable, interpretable, and privacy-preserving DR detection systems. Continuous improvements in optimization strategies, data augmentation

methods, and explainability algorithms are enhancing the effectiveness of these frameworks and supporting their seamless integration into real-world clinical workflows.

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